T12 Room Transitions

CRE132 worksheet

What you'll make

Two ways to move between areas. First, **rooms** on one map: the camera stays inside the meadow you're in, and when you walk through the gap in the fence it glides across to the next one. Second, a **building**: walk into the house's door, the screen fades, and you're inside, in a different scene. Walk back out and you arrive just outside the house, not back at the start of the level.

You'll use C02's Cinemachine camera and confiner, and S04's **SceneLoader** for the fade. The new idea is a **static** variable: a value that belongs to a class rather than to an object, so it survives a scene change.



Figure 1: The finished village: two meadows joined by a gap in the fence, with a house in the east room.

Before you start

1. Unzip `CRE132_T12_RoomTransitions_Starter.zip`. In Unity Hub, choose **Add > Add project from disk** and pick the `CRE132_T12_RoomTransitions_Starter` folder. Open it with Unity **6000.3.11f1**.
2. This sample has two scenes in `Assets/_game/Scenes/TopDown/`: **T12_RoomTransitions** (the village) and **T12_RoomTransitions_House** (inside the house). Both are already in the build list (**File > Build Profiles**), which a scene must be before it can be loaded by name. Open **T12_RoomTransitions**.
3. Press **Play** and walk around (WASD or arrows). Walking into the house's door does nothing yet.

Look under **Rooms** in the Hierarchy. **Room West** and **Room East** each have a **Box Collider 2D** (**Is Trigger** ticked) covering that meadow, and a child Cinemachine Camera (**CM Room West**, **CM**

Room East) whose **Cinemachine Confiner 2D** uses that same box as its **Bounding Shape 2D** (C02). Both cameras have **Priority** 0, so nothing tells the camera which room you're in. You'll fix that first.

> **Making a script:** in the Project window, go to the folder, right-click and choose **Create > Scripting > MonoBehaviour Script**. Type the name straight away and press Enter. The name must match the class name in the code exactly, capital letters included.

1. A camera per room: CameraRoom

Make a folder `Assets/_game/Scripts/T12_RoomTransitions` and create `CameraRoom` in it:

```
// One room of a bigger map, with its own Cinemachine camera kept inside the room by a
// Confiner 2D. While the player's feet are inside this room's box, its camera gets the
// higher priority. The Cinemachine Brain then blends smoothly from the old room's camera
// to this one (its Default Blend sets how long that takes and how it eases in and out).
using Unity.Cinemachine;
using UnityEngine;

public class CameraRoom : MonoBehaviour
{
    [SerializeField] private CinemachineCamera roomCamera;

    private Collider2D roomShape;
    private Transform player;

    private void Awake()
    {
        roomShape = GetComponent<Collider2D>();
        player = GameObject.FindWithTag("Player").transform;
    }

    private void Update()
    {
        if (player == null)
        {
            return;
        }

        // OverlapPoint asks "is this point inside the collider?". The feet are only ever
        // in one room, so exactly one room camera wins, even when the player stands in a
        // doorway.
        if (roomShape.OverlapPoint(player.position))
        {
            roomCamera.Priority = 10;
        }
        else
        {
            roomCamera.Priority = 0;
        }
    }
}
```

- **The Cinemachine Brain** (on the **Main Camera**) always shows the Cinemachine Camera with the highest **Priority**. The room you're in gets 10, every other room 0.
- **`OverlapPoint`** checks one point, the player's position (their feet). A point can only be in one box, so exactly one room wins, even in the gap between two rooms. Trigger enter and exit messages would get muddled there, with the player touching both boxes at once.

- **The smooth move is a blend.** When the winning camera changes, the Brain blends from the old one to the new one. Its **Default Blend** on the Main Camera is **Ease In Out**, **1** second.

Add it to **Rooms > Room West** and drag its child **CM Room West** into **Room Camera**. Do the same on **Room East** with **CM Room East**.

Test it: press Play and walk through the gap in the fence. The camera glides across and stays inside the new room. Select **Main Camera**, set the Brain's **Default Blend** to **Cut** and try again: the view jumps instead. Set it back to **Ease In Out**.

> **Rooms must be at least as big as the camera's view** (20 × 11.25 units here), or the confiner can't fit the view inside them. Both room boxes are 14 units tall and over 23 wide.

2. Where to arrive: SpawnPoint

Now the house. Going into another scene needs two things: which scene to load, and where to stand when you get there. The spot to arrive at is a **SpawnPoint**. Create `SpawnPoint` in the `T12_RoomTransitions` folder:

```
// A place the player can arrive at when coming from another scene (the spot just outside
// a door). When a scene starts, the spawn point whose name matches NextSpawnName moves
// the player to itself.
using Unity.Cinemachine;
using UnityEngine;

public class SpawnPoint : MonoBehaviour
{
    // Static: it belongs to the class, not to one object, so it keeps its value while one
    // scene is unloaded and the next is loaded. SpawnDoor sets it just before changing
    // scene.
    public static string NextSpawnName = "";

    [SerializeField] private string spawnName = "Front Door";

    private void Start()
    {
        if (spawnName != NextSpawnName)
        {
            return;
        }

        GameObject player = GameObject.FindWithTag("Player");
        player.GetComponent<Rigidbody2D>().position = transform.position;
        player.transform.position = transform.position;
        NextSpawnName = "";

        // Without this the cameras would glide over from where the scene was saved.
        // Telling each Cinemachine camera its last position is no longer valid makes it
        // jump straight to the player.
        foreach (CinemachineCamera followCamera in
            FindObjectsByType<CinemachineCamera>(FindObjectsSortMode.None))
        {
            followCamera.PreviousStateIsValid = false;
        }
    }
}
```

- **`static`** means `NextSpawnName` belongs to the `SpawnPoint` class itself, not to any one spawn point. When the old scene is unloaded, its objects go, but the class's static value stays. That's how a little information gets from one scene to the next.
- **Every spawn point checks the name in `Start`.** Only the one that matches moves the player, then it clears the name so it isn't used again by accident.
- **`PreviousStateIsValid = false`** tells each camera to forget where it was, so it snaps to the player instead of gliding across the map.

Add it to the two spawn points that are already in the scenes:

- In **T12_RoomTransitions**, select **Spawn Point (Front Door)** and add **SpawnPoint**. Leave **Spawn Name** as `Front Door`.
- Save, then open **T12_RoomTransitions_House**. Select **Spawn Point (Inside Door)**, add **SpawnPoint** and set **Spawn Name** to `Inside Door`.

Each spawn point is a step away from its door. If you arrived standing in the door's trigger, you'd be sent straight back.

3. A door to another scene: SpawnDoor

Create `SpawnDoor` in the same folder. It needs `SpawnPoint.NextSpawnName`, which is why you made SpawnPoint first.

```
// S04's SceneDoor, plus where to arrive. Before loading the other scene it writes down
// the name of the spawn point to use there; that scene's SpawnPoints read it and the
// matching one moves the player.
using UnityEngine;

public class SpawnDoor : MonoBehaviour
{
    [SerializeField] private string sceneToLoad;
    // The name of a SpawnPoint in the scene being loaded.
    [SerializeField] private string arriveAt;

    private void OnTriggerEnter2D(Collider2D other)
    {
        if (other.CompareTag("Player"))
        {
            SpawnPoint.NextSpawnName = arriveAt;
            SceneLoader.Instance.LoadScene(sceneToLoad);
        }
    }
}
```

- **`SpawnPoint.NextSpawnName = arriveAt`** uses the class name, not an object, because the value is static.
- **`SceneLoader.Instance`** is S04's loader. There is one of it, it survives scene loads, and it fades the screen while the next scene loads. Both scenes already have a **Scene Loader**.

Add it to the two door triggers. The names must be typed exactly, spaces and capitals included:

- In **T12_RoomTransitions_House**, select **Way Out** (the trigger at the bottom of the room) and add **SpawnDoor**. **Scene To Load** `T12_RoomTransitions`, **Arrive At** `Front Door`. Save.
- In **T12_RoomTransitions**, select **House > Door** (the small trigger in front of the door) and add **SpawnDoor**. **Scene To Load** `T12_RoomTransitions_House`, **Arrive At** `Inside Door`. Save.

Test it: play the village scene. Walk into the house's door: the screen fades and you're inside, by the door. Walk out through the gap at the bottom: you're back in the village, just outside the house, and the camera is already in the east room.

Check your work

- ☐ The camera stays inside whichever meadow you're in, and glides across when you go through the gap.
- ☐ The house door fades to the inside, and you arrive by the door.
- ☐ Leaving the house puts you just outside it, not at the start of the level, with no camera glide on arrival.
- ☐ You never get bounced straight back through a door.
- ☐ Compare with `CRE132_T12_RoomTransitions_Finished` if something behaves differently.

Key points

- **One camera per room, and the room you're in gets the highest priority.** The Brain does the smooth move between them.
- **Moving between scenes needs two things:** which scene to load, and where to stand when you get there.
- **A static variable** belongs to the class, not to an object, so it's a simple way to pass a little information from one scene to the next.

If something goes wrong

- **The camera doesn't switch rooms:** a **Room Camera** slot is empty or holds the other room's camera.
- **NullReferenceException in CameraRoom's Awake:** nothing in the scene is tagged **Player**.
- **"Scene couldn't be loaded" in the Console: Scene To Load** is misspelt, or the scene isn't in the build list.
- **You go through the door but arrive where the scene was saved: Arrive At** and **Spawn Name** don't match exactly. Check spaces and capitals.
- **NullReferenceException in SpawnDoor:** there's no **Scene Loader** in the scene you started from.
- **You arrive inside and are sent straight back out:** the spawn point is inside the door's trigger. Move it a step away.

Try it

1. Change the Brain's **Default Blend**: a faster **Ease In Out**, or a different style.
2. Add a third room to the north, with its own box, camera and gap in the fence. Make it at least 20 × 11.25 units.
3. A second house that uses the same interior scene. Which names do you need to change so you come out of the right house?
4. Harder: keep the player's health between scenes, using S02's GameManager.